

Regulative synthesis of capsular polysaccharides in the pathogenesis of *Streptococcus suis*

Reviewed Preprint

v2 • February 26, 2025

Revised by authors

Reviewed Preprint

v1 • October 30, 2024

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eLife Assessment

This **useful** study uses a model of *Streptococcus suis* (a pig pathogen) infection in mice using an intranasal route, the natural route of infection ignored in most of the literature. The study aims to understand how capsular polysaccharides (CPS) contribute to neuropathology and virulence. The findings suggest that the olfactory route may lead to meningitis before bacteremia occurs and that CPS down-regulation may play a role in this process. However, the study remains **incomplete** as presented.

<https://doi.org/10.7554/eLife.101760.2.sa2>

Abstract

Streptococcus suis (*S. suis*) is an important zoonotic pathogen causing substantial economic losses in the swine industry. *S. suis* serotype 2 (SS2) is often isolated from the diseased. *S. suis* expresses capsular polysaccharide (CPS), a virulence factor crucial for their survival in the blood. However, the role of CPS in the pathogenesis of *S. suis* is incomplete. Here, we showed a dynamic regulation of CPS in *S. suis* pathogenesis. In a mouse infection model, an SS2 strain (05ZYH33) was detected in the nasal-associated lymphoid tissue (NALT) and cerebrospinal fluid (CSF) as early as 30 min after intranasal inoculation without bacteremia. Histological analysis revealed that 05ZYH33 in the nasal cavity invaded the olfactory epithelium, resulting in early brain inflammation. Transmission electron microscopy showed that 05ZYH33 isolated from NALT and CSF at early infection time had a thin layer of CPS, and those detected in the blood 5 h post- inoculation showed a much thicker CPS. In addition, adoptive transfer of anti-CPS restricted 05ZYH33 in the blood but not in NALT or CSF. However, an antiserum directed to multiple non-CPS virulence factors (anti-V5) efficiently inhibited 05ZYH33 in NALT, CSF, and blood. Thus, 05ZYH33 colonizes NALT more efficiently without CPS and subsequently invades the meninges through the olfactory nerve system. These findings provide valuable information for the treatment of *S. suis* infection and the development of vaccines across serotypes of *S. suis* by targeting CPS-independent immunity.

Introduction

S. suis is a zoonotic pathogen in the porcine industry. Thirty-three serotypes of *S. suis* have been described to date based on the capsular polysaccharides (CPS). The *S. suis* serotype most frequently isolated from diseased swine and humans is serotype 2 (SS2). *S. suis* is commonly found in NALT of asymptomatic piglets. Under some unknown conditions, *S. suis* can reach and survive in the blood and invade the brain to cause meningitis and multi-organ dissemination. Infection of the central nervous system (CNS) is a cause of mortality in piglets with the disease. CPS is essential for survival of *S. suis* in the bloodstream and is recognized as an important virulence factor of this bacterial pathogen. However, the role of CPS in the pathogenesis of *S. suis* and the function of CPS-induced immunity in SS2 protection is controversial [1]. The opsonizing antibodies targeting CPS are protective [2, 3] but only partially protect against infection [4]. Furthermore, a CPS-glycoconjugated vaccine protected parenterally challenged piglets [5], but protection by anti-CPS has not been demonstrated in animals challenged through the intranasal route, which is a common natural infection route of *S. suis*.

Synthesis of CPS is enhanced in the bloodstream and reduced in the CNS [6, 7]. *In vitro* experiments revealed that the presence of CPS attenuated 05ZYH33 invasion of epithelial cells [8–10] and that unencapsulated mutants displayed increased rates of invasion of the blood–cerebrospinal fluid barrier (BCSFB) [8, 9]. These observations suggest that *S. suis* regulates CPS synthesis to fit hosts. However, despite the proposal of this hypothesis more than two decades ago, there is still no direct evidence of such encapsulation modulation for *S. suis* [11].

While CPS has been known for years and considered necessary for infection yet probably not sufficient, other surface-associated and secreted proteins of *S. suis* have received attention during the last years for their important virulent roles in the pathogenesis [3] and as vaccine candidates to raise effective immune response [15]. To cause disease, *S. suis* must colonize the respiratory mucosa, breach epithelial barriers, survive in the bloodstream, spread to different organs, and exaggerate inflammation. CPS and these virulent proteins contribute to the pathogenesis of the infection at specific steps.

Two pathways have been exploited by meningeal pathogens to access the CNS: penetrating the blood–brain barrier (BBB) or the BCSFB through the blood or directly accessing the CNS via the olfactory or trigeminal nerves [12]. Some human meningeal pathogens, such as *Streptococcus pneumoniae* and *Neisseria meningitidis*, can directly transmigrate from the nasal mucosa into the CNS through the olfactory nerve [12–14]. Similar to these meningeal pathogens, the predominant route of *S. suis* infection in pigs is through the upper respiratory tract (URT), which may provide the opportunity for *S. suis* to transmigrate directly to the CNS from the nasal cavity.

Studies on *S. suis* commonly use parenteral infection models that bypass colonization of the URT, thus this critical step of *S. suis* pathogenesis is largely unexplored. In the present study, mice were infected intranasally (i.n.) with 05ZYH33, and the infection process was monitored. We show that 05ZYH33 regulated the synthesis of CPS at different stages of infection and that 05ZYH33 migrated to the CNS from the nasal cavity independent of CPS before it entered the blood. Moreover, we revealed that immunity against non-CPS virulence factors of *S. suis* efficiently prevented nasal–CNS transmigration and subsequent progress of the infection.

Results

S. suis serotypes regulated CPS synthesis in different culture phases

A clinical isolate of *S. suis* serotype 2 wild type (05ZYH33) was cultured and collected in the lag, logarithmic (log), and stationary phases [optical density at 600 nm (OD600) of 0.20, 0.60, and 0.80, respectively] (Fig. 1A). The CPS of 05ZYH33 collected in different growth phases was quantitated based on hydrophobicity in a bacterial adhesion to hydrocarbon (BATH) assay. High surface hydrophobicity leads to high bacterial adherence to hydrocarbon (high BATH value) and reflects low levels of CPS on the surface of 05ZYH33. As shown in Fig. 1B, the CPS-deficient strain Δ 2BSS2 (05ZYH33 isogenic), used as a control, exhibited the highest BATH value ($69.5 \pm 6.51\%$). A similarly high BATH value ($62.5 \pm 9.73\%$) was found in the log phase of 05ZYH33. However, the BATH values in the lag and stationary phases of 05ZYH33 were substantially lower ($37.0 \pm 17\%$ and $15.6 \pm 9.2\%$, respectively). These data indicate that CPS of 05ZYH33 was synthesized dynamically *in vitro*, with the highest quantity in the stationary phase and the lowest in the log phase.

To determine the dynamic CPS synthesis in *S. suis* serotypes, four other serotypes of *S. suis* were cultured and collected in the log and stationary phases for CPS quantitation by BATH assays. As shown in Fig. 1C, all the serotype strains had high hydrophobicity at the log phase and low at the stationary phase. The similarity in CPS patterns of these strains suggests a shared regulation mechanism by *S. suis* serotypes.

05ZYH33 in different cultural stages exhibited altered abilities for host cell adherence and resistance to bacterial killing

05ZYH33 and Δ 2BSS2 were cultured and collected at the log and stationary phases to determine the effects of CPS on the binding of 05ZYH33 to epithelial cells. The adherence assay showed that while $16.5 \pm 5.2\%$ of Δ 2BSS2 was associated with epithelial cells (HEp-2), only $4.96 \pm 3.26\%$ of 05ZYH33 in the log phase and $2.05 \pm 1.58\%$ of 05ZYH33 in the stationary phase were associated with the cells (Fig. 2A). The bacterial binding was Δ 2BSS2 > 05ZYH33 in the log phase > 05ZYH33 in the stationary phase. Similar binding patterns of 05ZYH33 and Δ 2BSS2 were also observed in the assay using human brain microvascular endothelial cells (hBMECs), the major component of the BBB (Fig. 2B). These results support that CPS reduces adherence of the bacteria to host cells.

CPS of *S. suis* plays a critical role in the resistance to bacterial killing. 05ZYH33 in different growth phases were co-incubated with the blood from naïve mice to determine the effect of CPS on bacterial killing. After incubation for 3 h, surviving bacteria in the blood examined by colony-forming units (CFUs) showed that substantially more 05ZYH33 in the log and stationary phases were detected compared with Δ 2BSS2 in the matching growth phases. In addition, there were more surviving 05ZYH33 in the stationary phase compared with that in the log phase (Fig. 2C). These results suggest a correlation between lower CPS in the NALT and a greater capacity for cellular association, whereas elevated CPS levels in the blood are linked to improved resistance against bactericidal activity. However, the mechanisms behind these associations remain unknown.

The course of 05ZYH33 infection in mice following intranasal inoculation

The dynamic synthesis of CPS associated with differential adherence to host cells and resistance to bacterial killing *in vitro* suggested that 05ZYH33 regulates CPS synthesis for pathogenesis and survival in a host. To determine the role of CPS dynamic synthesis in 05ZYH33 pathogenesis, mice

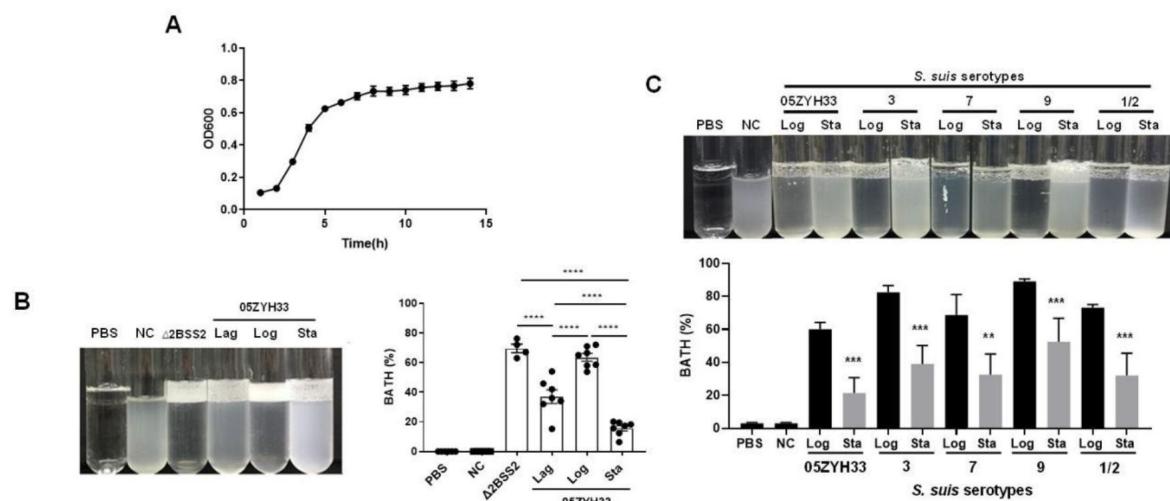


Fig. 1.

Monitoring the CPS content of *S. suis* serotypes in culture.

(A) The growth curve of 05ZYH33 in TSB-FBS culture was assessed by measuring the optical density at 600 nm (OD600) at specified time points. Data present the mean $\pm$ SD of three independent experiments ($n = 10$). (B) The hydrophobicity of 05ZYH33 during the lag, log, and stationary (Sta) phases was measured by BATH assays ($n = 4$ to 7). (C) The hydrophobicity of indicated serotype strains in the log and stationary phases was measured by BATH assays. A representative photo of the BATH assay (upper panel) and BATH values (lower panel) ($n = 3$ to 6). Δ 2BSS2 refers to an isogenic strain lacking CPS; NC represents a 05ZYH33 suspension without hydrocarbon. **, $P < 0.01$. ***, $P < 0.001$. ****, $P < 0.0001$.

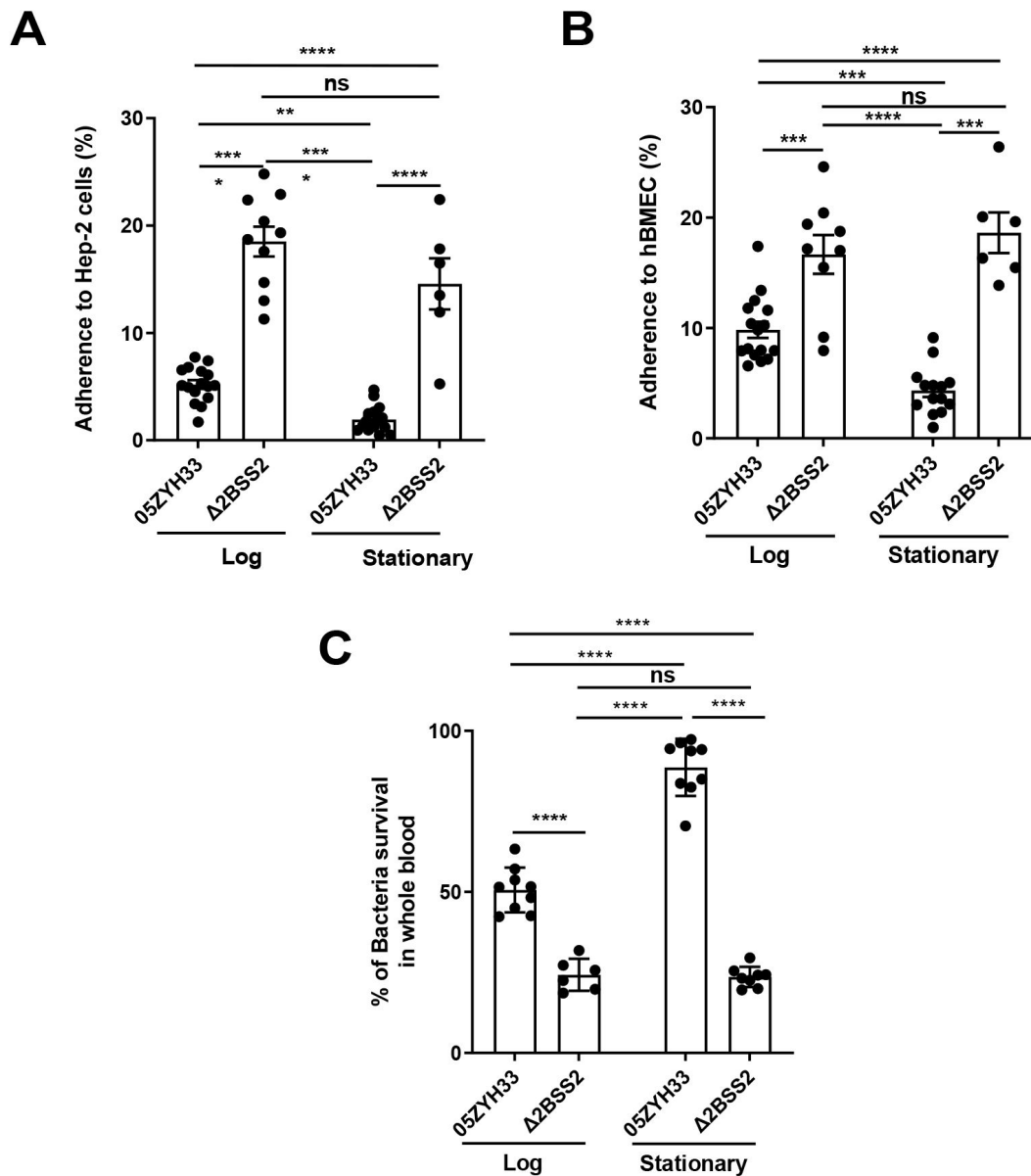


Fig. 2.

The adherence of 05ZYH33 to epithelial cells and its resistance to bacterial killing.

HEp-2 or HBMECs cells were inoculated with 05ZYH33 or Δ2BSS2 in either the log or stationary phase at an infection MOI of 10. (A) Adherence rate of *S. suis* to HEp-2 cells. (B) Adherence rate of *S. suis* to HBMECs. (C) *S. suis* strains in log and stationary phases were incubated with whole blood from naïve mice for 3 h, and the viability of *S. suis* was determined. Data are from three independent experiments and present as means ± SEM ($n = 6$ to 16). ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$; ns, not significant.

were i.n. inoculated with 05ZYH33 to simulate *S. suis* natural infection in pigs and CFUs in various bodily compartments were assessed at different points over 120 h. By 0.5 h after inoculation, CFUs of 05ZYH33 in the nasal-associated lymphoid tissue (NALT), a pig tonsil homologue and the major niche for *S. suis* in pigs, and the lungs were similar (5×10^5) and gradually reduced with time (Figs. 3A and 3B), suggesting that the CFUs in the lungs are from the nasal cavity. Up to 10^5 CFUs were also found in CSF samples by 0.5 h (Fig. 3C). The curves of CFUs in NALT, the lungs, and CSF showed a similar trend, which was a high level for a few hours followed by a gradual reduction with increasing days. In contrast, 05ZYH33 was not detected in the blood until 5.0 h after inoculation (Fig. 3D). As expected, CFUs in the spleen and heart at indicated times (Figs. 3E and 3F) were similar to those in the blood, indicating that they disseminated from the circulation. These findings suggested that 05ZYH33 in the nasal cavity can transmigrate to the lungs and CNS before bacteremia and disseminate to other organs after entering the blood. During the five days (120 h) following inoculation, the mice did not show significant disease symptoms, and their physical conditions were similar compared to non-infected mice (body weight, appetite, and activity). No obvious disease symptoms might be due to the bacteria cleared relatively quickly.

The CPS of 05ZYH33 was synthesized differentially in stages of infection in mice

05ZYH33 was isolated from the bodily compartments of mice 12 h after inoculation. The 12 h of infection allows the detection of the bacteria in different body compartments of the infected mice. Following enrichment by culturing the isolates in TSB to an OD600 of 0.7 CPS synthesis *in vivo* was determined by BATH assay. The hydrophobicity of isolates from NALT and the CSF was lower compared with that of $\Delta 2BSS2$, but higher than that of the isolates from the blood (Fig. 4A). This indicates that 05ZYH33 downregulates CPS synthesis when it colonizes the nasopharyngeal mucosa or enters the CNS and upregulates CPS synthesis upon entering the blood. Transmission electron microscopy (TEM) was performed directly on 05ZYH33 isolated from mice without *in vitro* culture to confirm the BATH assay results. For cultured strains without mouse infection (Fig. 4Ba, b, and c), similar to $\Delta 2BSS2$ (Fig. 4Ba), in the log phase (Fig. 4Bb) there was no CPS observed on the dividing 05ZYH33 cell (left and larger cell) and only a thin and partial layer of CPS on the smaller dividing cell (right, smaller cell) was observed. Whereas, a thicker layer of CPS was observed on the surface of the bacteria in the stationary phase (Fig. 4Bc). For 05ZYH33 isolated directly from infected mice (Fig. 4Bd, e, and f), the CPS was much thicker, denser, and uniformly distributed in the blood isolates (Fig. 4Bf) compared with that of the NALT isolates where it was thinner and much sparser (Fig. 4Bd). In contrast, the outermost layer of the isolates from the CSF was highly dense without an apparent CPS layer (Fig. 4Be). These findings support that CPS downregulation is associated with efficient colonization and upregulation with increased resistance to bacterial killing in the blood.

An antiserum against non-CPS virulence factors of *S. suis* reduced 05ZYH33 colonization in NALT, CSF, and lungs

We have previously demonstrated that immunization with multiple non-CPS virulence factors of *S. suis* (named V5) prevents 05ZYH33 colonization of the nasal mucosa and neuropathological symptoms in mice [15]. We hypothesized that the serum against V5 would inhibit the nasal–CNS transmigration of 05ZYH33. Mice were i.n. infused with anti-V5 or anti-CPS serum, inoculated i.n. with 05ZYH33 6 h later, then euthanized 12 h post-inoculation (Fig. 5A) to detect the bacteria in different body compartments of the infected mice. In mice that received antisera through the i.n. route, CFUs reduced in all examined compartments of anti-V5 recipients but only reduced in the lungs and blood of anti-CPS recipients (Fig. 5B). These results indicated that 05ZYH33 survival in the blood depends on CPS and nasopharyngeal colonization and nasal–CNS transmigration are more dependent on non-CPS molecules of 05ZYH33. In addition, when mice received either anti-V5 or anti-CPS through the intravenous (i.v.) route, CFUs were reduced in the lungs and blood but not

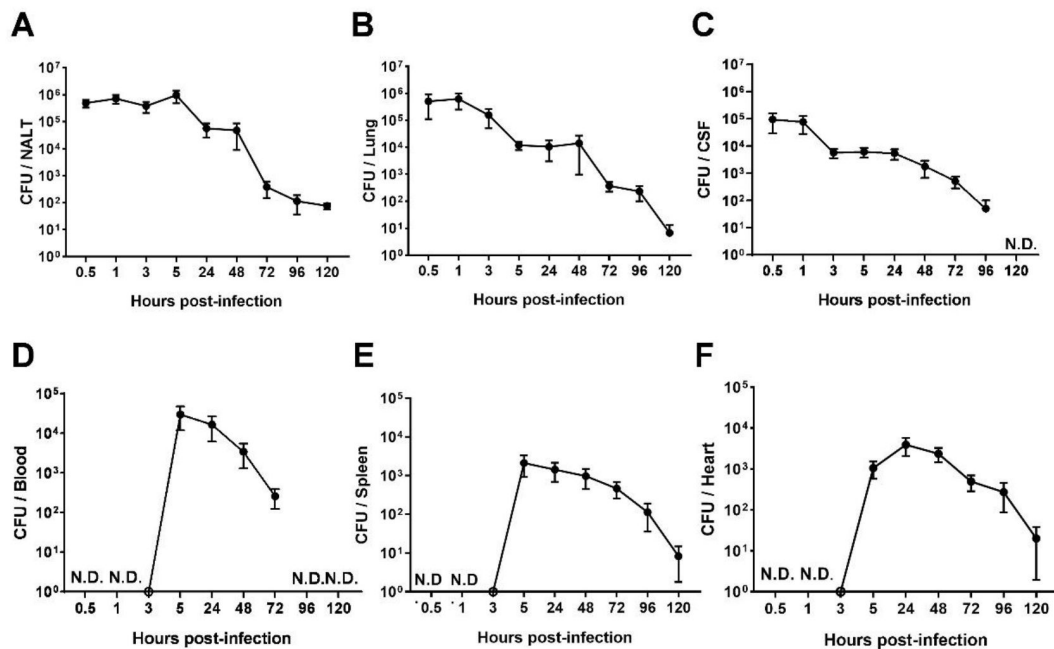


Fig. 3.

The intranasal infection course in mice.

Mice were intranasally inoculated with 05ZYH33. Total CFUs in various bodily compartments were assessed at the indicated time points after inoculation, (A) NALT, (B) the CSF, (C) the lung, (D) the blood, (E) the spleen, and (F) the heart. The data presented are derived from two independent experiments and presented as means $\pm$ SEM ($n = 6$). N.D., not detected.

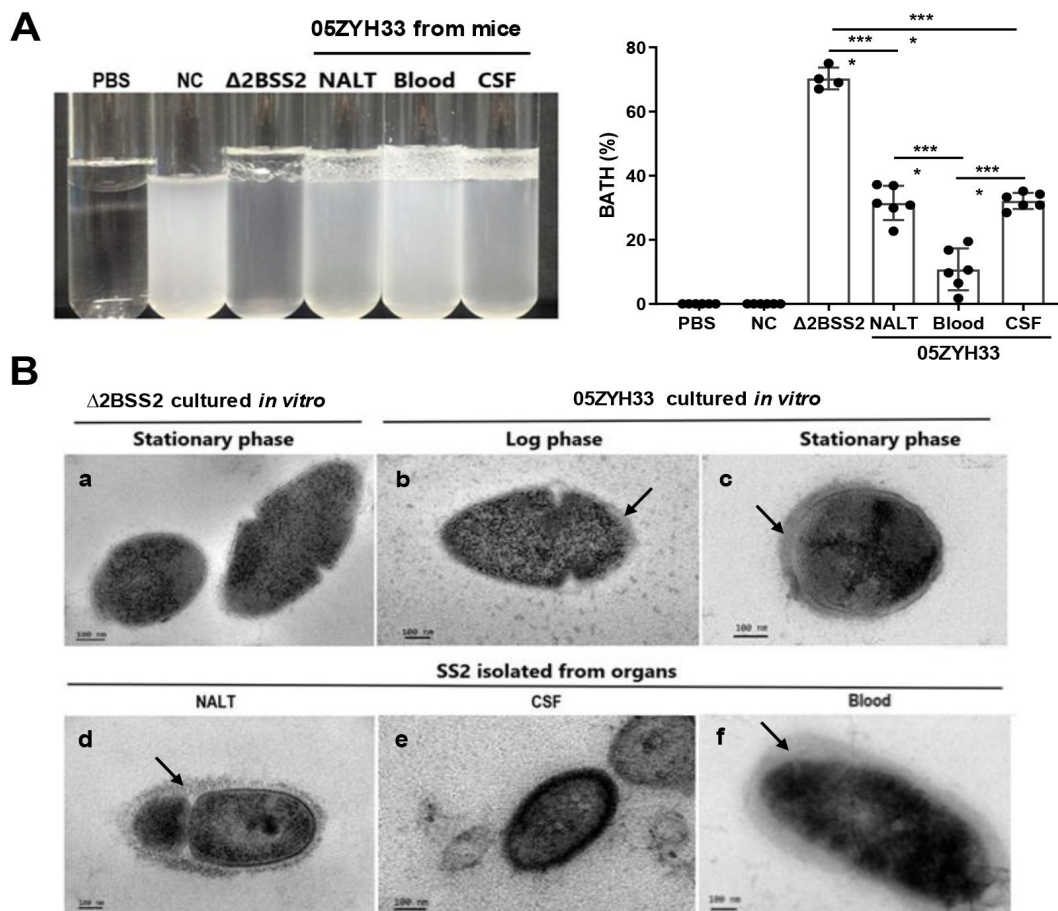


Fig. 4.

The CPS of 05ZYH33 isolated from various bodily compartment following infection.

05ZYH33 was intranasally inoculated in mice. Twelve hour after inoculation, CFUs in NALT, the CSF, and the blood were assessed for (A) hydrophobicity of bacterial cells by BATH assay and (B) CPS by TEM. Arrows indicate the CPS layers. Scale bar, 100 nm. Data in (A) are from two independent experiments and present as means $\pm$ SEM ($n = 4$ to 6). NC, 05ZYH33 suspension without hydrocarbon. **** $P < 0.0001$.

in NALT and the CSF (**Fig. 5C**), compared with mice received naïve serum through the i.v. route. These observations suggest that systemic immunity could efficiently protect against infection in the lungs and blood but not against colonization in the nasal mucosal and CNS, at least at the early infection time.

The simultaneous 05ZYH33 colonization in NALT and CNS support a nasal–CSF transmigration. To exclude the possibility that CFUs in the CNS at 12 h post-infection is disseminated from the circulation, mice received antisera via the i.n. route and were inoculated with 05ZYH33 as before, then CFUs were determined 1 h post-inoculation before the bacteria entered the blood (**Fig. 5A**). As shown in **Fig. 5D**, CFUs in NALT and the CSF of anti-CPS recipients were similar to those in naïve-serum recipients; however, those in NALT and the CSF of anti-V5 recipients were 10-fold lower compared with those of the naïve-serum recipients. To confirm that the nasal–CNS transmigration is independent of CPS, mice received naïve serum or anti-V5 via the i.n. route, then were inoculated with $\Delta 2BSS2$. One hour later, we found 4.5×10^5 CFUs in NALT and 2.1×10^5 CFUs in the CFS of naïve-serum recipients. However, CFUs were 10-fold and 100-fold lower in NALT and the CSF, respectively, of anti-V5 recipients compared with those in naïve-serum recipients (**Fig. 5E**). These results demonstrate that 05ZYH33 in the CNS in the early infection is from the nasal cavity independent of CPS.

The olfactory nerve is the invasion route of 05ZYH33 for nasal–CNS transmigration

The olfactory nerve is a port of entry to the meninges for some human meningeal pathogens and is the likely the route of the early CNS dissemination by 05ZYH33. One hour after infection, the mice were sacrificed, and their heads were sectioned and stained for histological analysis. **Figure 6A** presents a histological overview of the anatomical structures the distal NC and OB regions, while **Figure 6B** illustrates the locations of the anatomical compartments in the area. The tissue section showed that compared with uninfected mice (**Fig. 6A, a, c, and e**), Scattered and grouped bacteria were observed on the surface of and in the olfactory epithelium (OE) (**Fig. 6A, b, and d**), orange signal, arrowheads), accompanied by severely destroyed tissue (indicated by the loss of immunofluorescence signal in the area) (**Fig. 6A, d**). Large groups of bacteria were observed within the olfactory bulb (OB) (**Fig. 6A, f**). These observations suggest that 05ZYH33 in the nasal cavity could invade the olfactory epithelium through the sensory neurons protruding into the NC and subsequently enter the brain along the sensory neurons at the cribriform plate. Histological analysis of the brain in 05ZYH33-infected mice revealed that inflammatory cells were infiltrated in the distal part of the brain tissue with surrounding edema at 1 h post-inoculation (**Fig. 6C, c and d**). The pathological changes were more severe 9 days after 05ZYH33 inoculation (**Fig. 6C, e and f**). Since CSF flows into the subarachnoid space of the spinal cord and around the brain [16, 17] the distal inflammation is consistent with the brain's anatomical structure. Thus, nasal–CNS transmigration is an efficient route of *S. suis*-mediated CNS infection.

Mice were pretreated with acetic acid to increase the sensitivity to 05ZYH33 infection [15]. To exclude that nasal–CNS transmigration results from irritation of the nasopharyngeal mucosa by acetic acid, mice were infected without acid treatment (pretreated with PBS). One hour post-inoculation CFUs were found in the CSF of the mice even though the numbers were lower than those of acidic acid-treated mice (**Fig. 6D**). CFUs were also detected in the CSF when mice were infected with $\Delta 2BSS2$ without acid treatment (**Fig. 6E**). These results indicated that nasal–CSF transmission takes place without acidic acid and independent of CPS and that irritation of the nasopharyngeal mucosa facilitates nasal–CNS transmission.

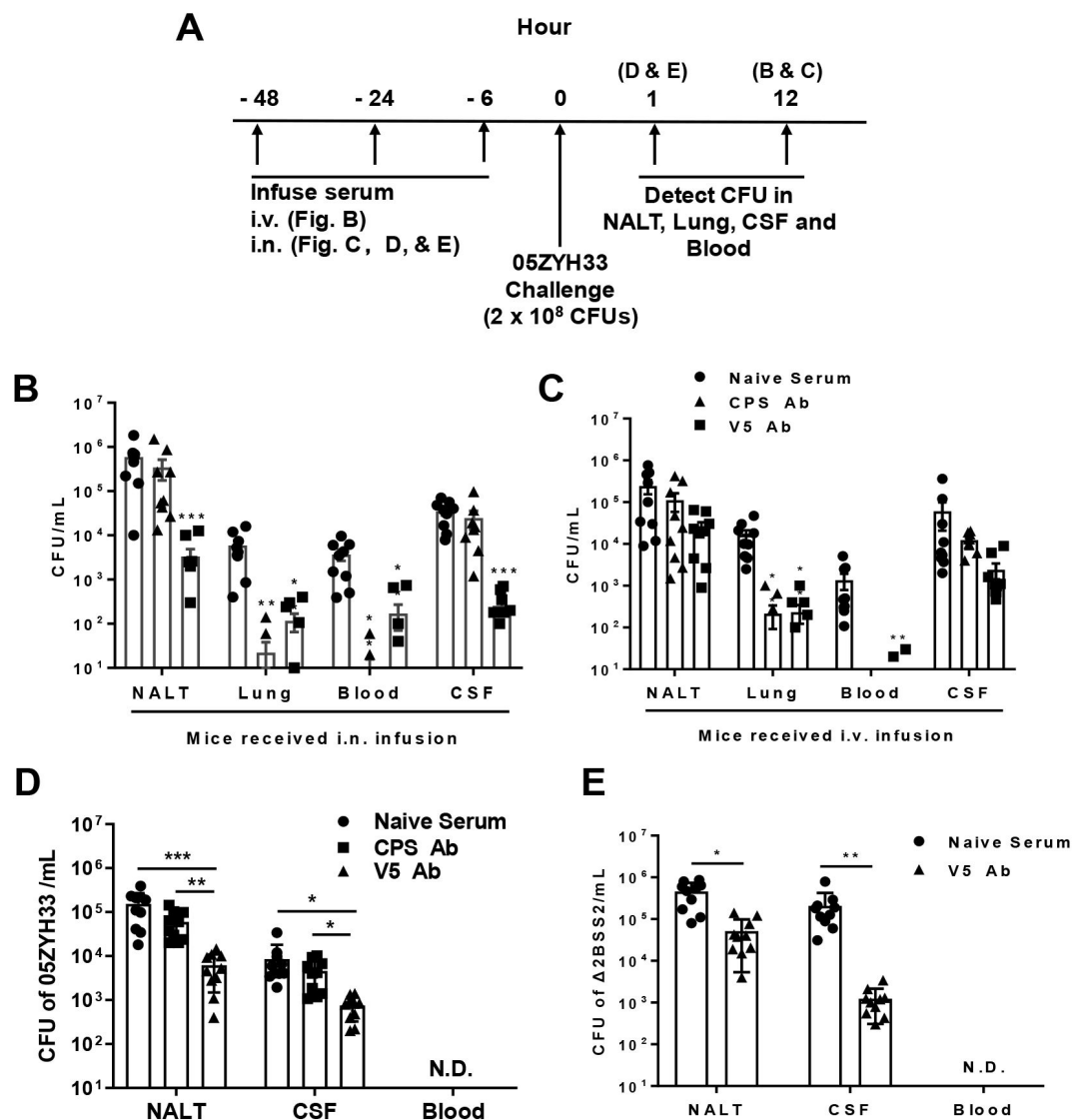


Fig. 5.

The role of anti-CPS or anti-V5 serum in 05ZYH33 infection.

(A) Schematic illustration of passive transfer of antisera and infection in mice. (B) and (C) Anti-V5 or anti-CPS antiserum was transferred through the i.n. or i.v. route. Six hours later, mice were i.n. challenged with 05ZYH33, then euthanized 12 h post-challenge. CFUs of 05ZYH33 in NALT, the lungs, the blood, and the CSF were determined. (D) and (E) Mice were infused with anti-V5 or anti-CPS serum through the i.n. route, and inoculated i.n. with 05ZYH33 or $\Delta 2BSS2$ 6 h later. One hour later, mice were euthanized, and CFUs in NALT, the blood, and the CSF were determined. Data are the mean $\pm$ SEM of three independent experiments ($n = 9 - 10$). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

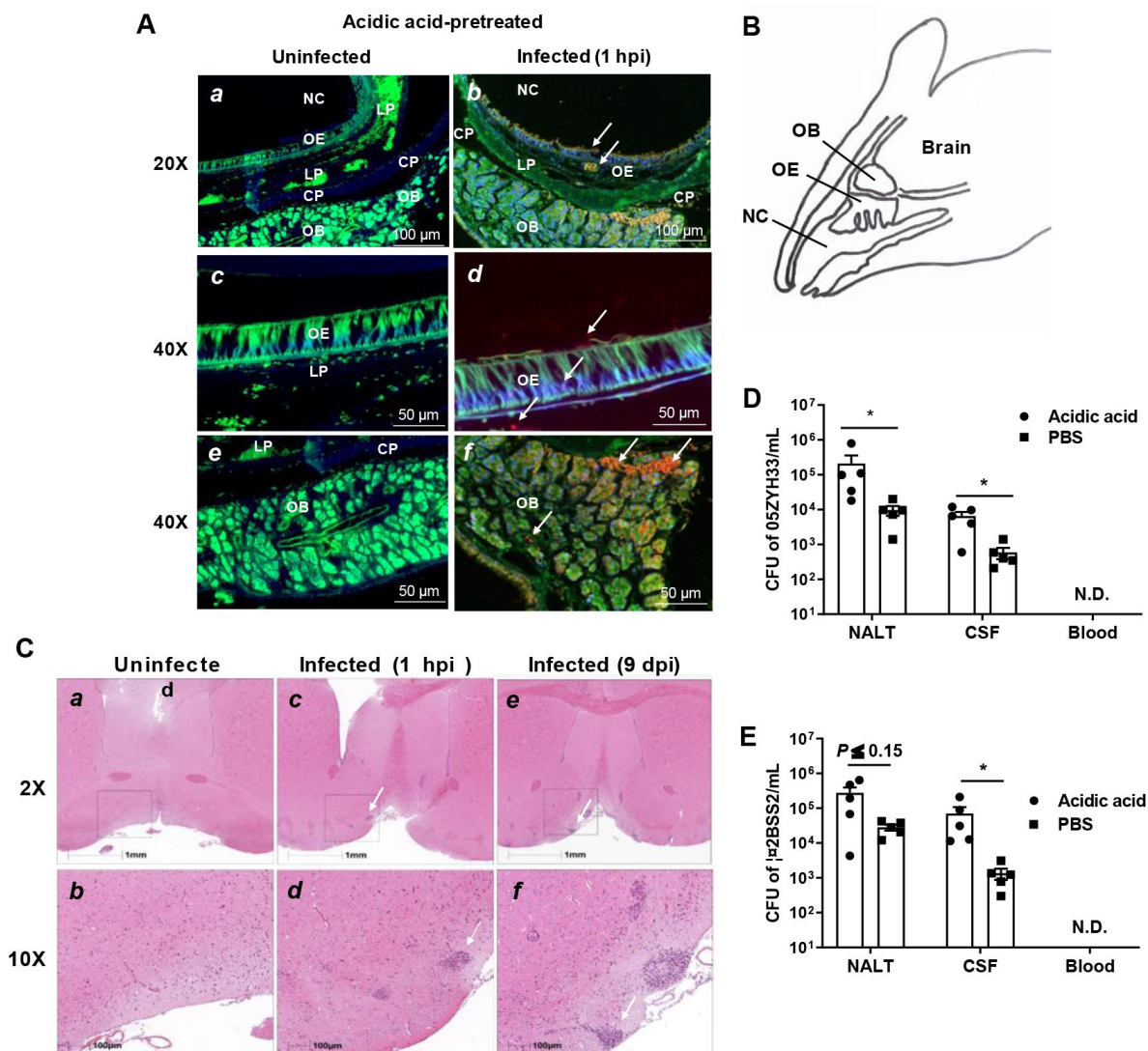


Fig. 6.

Detection of the presence of 05ZYH33 in the tissue of olfactory nerve area.

Mice were inoculated with acetic acid through the nostril and one hour later infected i.n. with 05ZYH33. (A) Sagittal views of the olfactory system. Sections of the distal nasal cavity and olfactory epithelium from uninfected (*a*, *c*, and *e*) and 05ZYH33-infected (*b*, *d*, and *f*) mice. Red or orange, 05ZYH33; green, the neuronal marker β -tubulin III; blue, DNA. NC, Nasal cavity; OE, olfactory epithelium; LP, lamina propria; CP, cribriform plate; OB, olfactory bulb. (B) A draw of the sagittal plane of the rodent nose elucidates the compartments of the olfactory bulb (OB), olfactory epithelium (OE), nasal cavity (NC), and the brain. (C) One hour or nine days after 05ZYH33 infection, the brain of mice were sectioned for H&E staining. Arrows indicate infiltrated inflammatory cells. (D) and (E) Mice were inoculated with acetic acid or PBS and one hour later infected i.n. with 05ZYH33 or Δ 2BSS2. (D) CFUs of 05ZYH33 or (E) Δ 2BSS2 in NALT, the CSF, and blood were determined one h after infection.

Discussion

CPS is an important virulence factor of *S. suis*, but its role in *S. suis* pathogenesis is controversial. Most animal models in studies of *S. suis* employ the intraperitoneal (i.p.) or i.v. route for infection, which bypass the nasopharynx, the natural infection route of *S. suis*.

In this study, CPS thickness of *S. suis* SS2 strain 05ZYH33 was monitored in an i.n. infection mouse model. The results indicate that *S. suis* CPS can be dynamically synthesized during the development of the bacterial infection. We showed that the downregulation of CPS synthesis by 05ZYH33 is associated with increased nasopharyngeal colonization and early nasal–CNS transmigration before bacteremia. This is consistent with previous studies that showed the upregulation of CPS is related to increased resistance to bacterial killing later in the blood. The study also indicates that CPS-independent immunity can prevent nasal–CNS transmigration and the progression of 05ZYH33 infection.

Clinical CNS symptoms in pigs are often preceded by a phase of bacteremia [16, 17], indicating that *S. suis* meningitis results from the dissemination of bacteria from the blood through the BBB or the blood–CSF barrier. Our study revealed that, like some human meningeal pathogens, the olfactory nerve epithelium provides an alternative route for 05ZYH33 invasion to the CNS. Swine influenza virus (SIV) infection was reported to promote *S. suis* invasion of deeper tissues by damaging epithelial cells [18–20]. Nasopharyngeal inflammation by co-infected *S. suis* and SIV could cause tissue damage to the olfactory epithelium, which may exacerbate nasal–CNS entry. As the mouse is not a natural host of *S. suis* and is insensitive to the infection, a relatively high dosage is needed for optimal bacterial retention in NALT, which can lead bacteria into the lungs. Thus, early 05ZYH33 colonization in the lungs of this model may not reflect the state of *S. suis* infections in pigs. The early 05ZYH33 colonization in NALT and the CSF with similar dynamics suggests that meningitis could be early manifestations of *S. suis* disease before multi-organ dissemination through the blood. The findings in the study indicate that restriction of *S. suis* proliferation in NALT is critical to prevent the severity and lethality of *S. suis* disease.

We noticed that the outermost layer of the isolates from the CSF was highly dense without an apparent CPS layer (**Fig. 4Be**), suggesting that other membrane components of the CSF isolates are also changed besides CPS. It is unclear whether this change is related to CSF colonization and what molecules have changed. Although CPS plays an important role in *S. suis* survival in the blood, it is not the only component required for the event [21, 22]. Similar to that observed with anti-CPS, infusion of anti-V5 through the i.v. route reduced 05ZYH33 in the blood and lungs, indicating that blocking CPS-independent mechanisms can restrain the progression of 05ZYH33 infection and suggests that non-CPS virulence factors, such as those in V5, are required for 05ZYH33 survival in the blood and lungs. Furthermore, although anti-CPS promoted 05ZYH33 clearance in the blood, it did not prevent 05ZYH33 colonization of the nasal cavity and invasion of the CNS. However, anti-V5 reduced 05ZYH33 in all examined compartments when administered via the i.n. route, supporting that inhibition of nasal colonization by CPS-independent immunity is critical to prevent the progression of the disease. Moreover, the nasal–CNS route may not be the main pathway for clinical meningitis compared to the blood–CNS route. However, mucosal blockage of nasal colonization reduces CFU in the CSF more effectively than systemic antibody, indicating reduction of nasal colonization at early time would significantly prevent the bacteria from entering the CNS through the early nasal and later blood route.

In summary, this study supports an updated *S. suis* infection model. Briefly, 05ZYH33 downregulates CPS to colonize the URT mucosa, where it can transmigrate to the CNS through the olfactory epithelium early in the infection process. 05ZYH33 upregulates CPS to avoid bacterial killing when it enters the blood, leading to multi-organ dissemination. The regulation of CPS at

different stages of 05ZYH33 pathogenesis and the protection offered by CPS-independent immunity provide valuable information for clinical treatment and development of vaccines across serotypes of *S. suis*.

Materials and methods

Ethics statement

This study was performed in strict accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the IMCAS (Institute of Microbiology, Chinese Academy of Sciences) Ethics Committee. The protocol was approved by the Committee on the Ethics of Animal Experiments of IMCAS (permit no. APIMCAS2019004). Mice were bred under specific pathogen-free (SPF) conditions in the laboratory animal facility at IMCAS. All animal experiments were conducted under isoflurane anesthesia, and all efforts were made to minimize suffering.

Bacterial strains and growth conditions

S. suis strain SS2 05ZYH33 (GenBank no. for complete genome sequence: CP000407.1) was isolated from a fatal human case of streptococcal toxic shock-like syndrome (STSLs) in 2005 in Sichuan province, China. A capsular mutant of *S. suis* 05ZYH33 (Δ 2BSS2) was obtained from Prof. Jia-Qi Tang (General Hospital of the Eastern Theater of the People's Liberation Army, China) [23]. *S. suis* serotypes SS3, SS7, SS9, and SS1/2 were gifts from Dr. Shengbo Cao (National Key Laboratory of Agricultural Microbiology, Huazhong Agricultural University, Wuhan, China). The information of these strains is listed in Supplementary Table 1. All *S. suis* strains were cultured in tryptic soy broth (TSB) containing 5% newborn bovine serum (FBS) or on tryptic soy agar (TSA) supplemented with 5% FBS at 37°C and 5% CO₂. The cultured strains were washed and resuspended in phosphate-buffered saline (PBS). CFUs were verified by plating on blood agar.

Growth curve of 05ZYH33

05ZYH33 was grown in TSB-FBS medium to an OD₆₀₀ of approximately 0.4 and then diluted 1000 times with fresh medium. Next, 200 μ L of the bacterial suspension was added in quadruplicate to a 100-well honeycomb plate and cultured at 37°C and 5% CO₂ for 16 h. The OD₆₀₀ of each well was measured at different times using a CMax Plus plate reader (Molecular Devices, San Jose, CA, USA).

Determination of CFUs in various bodily compartments

Mice were infected i.n. with 05ZYH33 (stationary phase). Then at different time points, the lung, spleen, heart, and NALT were taken from mice, and the blood and CSF were collected. A single-cell suspension of each organ was prepared in 1-5 mL PBS. CFUs in the single cell suspension, the blood, and the CSF were determined by plating serially diluted samples on blood agar plates and expressed as total CFUs in each organ or in the blood and CFS by following formula: CFUs on the plate $\times$ dilution factor $\times$ total volume of a sample.

Bacterial adhesion to hydrocarbon (BATH) assay

The hydrophobicity of *S. suis* was assessed by a modified BATH assay [24]. Briefly, 05ZYH33 cells in cultures were collected, washed, and resuspended in PBS, adjusting the OD₆₀₀ of the suspensions to 2.0 ($\pm$ 0.2) with PBS. Four milliliters of each suspension were added to a tube containing 300 μ L xylene and 300 μ L hexadecane. A negative control comprised a 4 mL bacterial suspension without hydrocarbon. After vortexing for 20 s, the tubes were maintained at 37°C for 30 min. The OD₆₀₀ of the resulting lower aqueous layer and the tube contents before vortexing were measured. The ratio of the absorbance of the bacteria assay tubes (Ab) to the absorbance of the bacteria control (Ac) was calculated as adhesion to hydrocarbon (%) = $(Ac - Ab) / Ac \times 100$.

Bacterial adherence assay

The human laryngeal epithelial cells (HEp-2, CL-0104) were purchased from Procell Life Science & Technology Co., Ltd. Wuhan, Hubei Province and cultured according to the manufacturer's instruction. The human brain microvascular endothelial cells (hBMEC) were kindly provided by Prof. Xiang-Ru Wang in Huazhong Agricultural University, Wuhan, Hubei, China and cultured as described by Yang [25]. Adherence of 05ZYH33 or Δ2BSS2 in log and stationary phases was performed as described previously with slight modifications [26]. Cells were cultured in 24-well plates to 80–90% confluence ($\sim 1.2 \times 10^5$ cells/well), then starved overnight in DMEM without serum and antibiotics. Cell monolayers were washed with PBS and overlaid with DMEM medium with 10% FBS (500 μ L/well) containing 05ZYH33 or Δ2BSS2 (MOI = 10). After incubation for 2 h at 37°C and 5% CO₂, unbound bacteria were removed, and the cells were lysed with 0.025% Triton X-100. CFUs in samples of diluted cell lysate were determined on blood agar plates and expressed as the percentage of adherence: (CFUs in wells with cells / CFUs in wells without cells) $\times$ 100.

Bactericidal assay

Bactericidal assays were performed as described by Pian et al. [27] with slight modifications. Briefly, 1×10^5 05ZYH33 or Δ2BSS2 in log and stationary phases were added to whole blood from naïve mice. The mixtures were rotated for 3 h at 37°C and 5% CO₂. A set of the same bacterial samples was added to naïve serum to determine bacterial numbers in the 3 h incubation without killing. Viable bacterial counts were determined by plating diluted samples on blood agar. The anti-killing capacity was calculated as (CFU in blood / CFU in serum) $\times$ 100%.

Mouse infection

C57BL/6J Cnc wild-type mice (aged 6 to 7 weeks) were purchased from Vital River Laboratory Animal Center (Beijing, China) and randomly assigned to groups. Mice were pretreated with 5 μ L of 1% acetic acid (pH 4.0) or PBS per nostril 1 h before intranasal infection [15]. After further anesthesia, mice were inoculated intranasally with 05ZYH33 (stationary phase, 2×10^8 CFU / 10 μ L). 5 μ L of the inoculum was delivered intranasally to each nostril (total 10 μ L). This volume allows optional bacterial retention in NALT.

Specimen collection and preparation

Mice were euthanized, then blood samples were collected from the heart, and the lungs, heart, spleen, and NALT were removed and washed with PBS to eradicate blood contamination. Single-cell suspensions of the tissues were prepared by mechanical dissociation of tissue through a 100- μ m nylon mesh in 1 mL PBS. For CSF collection, the nose part was removed from the head along the line of the eyeballs after the collection of NALT tissue. The brain tissue was taken from the remaining part of the head and soaked in 1 mL PBS for 1 h after sufficient laceration to expose the subarachnoid space without mincing. This operation avoided damaging the integrity of brain tissue as much as possible. Supernatants of the single-cell suspension and the blood and CSF samples were used to detect CFU and examine the CPS.

Transmission electron microscopy (TEM)

S. suis cells were enriched by gradient centrifugation of homogenized NALT and samples of the blood and CSF of infected mice. After washing, the bacteria were fixed in 2% paraformaldehyde supplemented with 0.05% ruthenium red in 0.1 M phosphate buffer (pH 7.0) for 30 min at room temperature. The bacterial samples were further prepared and embedded in epoxy resin (Epon 812, Merck) as described by Bojarska et al. [28]. Ultrathin (70–80 nm) sections were cut, stained with 1.2 % uranyl acetate and 2.5 % lead citrate, and then examined with a JEM-1400 (120 kV) transmission electron microscope (JEOL, Tokyo, Japan). Images were taken with a Morada digital camera (Olympus SIS, Münster, Germany) at 10 M pix resolution.

Cloning and expression of recombinant V5 proteins

V5 contains five conserved virulence factors of *S. suis*. They are SrtA (amino acids 50 to 249), ScpCL (amino acids 36 to 535), MRP (amino acids 283 to 721), SLY (amino acids 28 to 497), and SspA (amino acids 41 to 852) and prepared as described by Xing X. et al. (15). Briefly, the genes of V5 were cloned from the genome of *S. suis* strain A7 (serotype 2) (GenBank accession number: NC_017622.1). The recombinant proteins of V5 were expressed in *E. coli* BL21 (DE3) and purified by immobilized metal ion affinity chromatography (IMAC) to reach 95% purity. Bacterial lipopolysaccharide was removed following purification (≤ 0.1 EU/ μ g). Protein concentrations were determined using the BCA Protein Assay Kit according to the manufacturer's instructions (TIANGEN, catalog number: PA115-02).

Preparation and purification of anti-CPS and anti-V5 antibodies

A female New Zealand white rabbit (SPF, 2.0–2.5 kg) was purchased from JinMuYang Laboratory Animal Center (Beijing, China) and kept in the SPF animal facility for 1 week. V5 (1 mg) was mixed with 1 mL incomplete Freund's adjuvant (Becton Dickinson, Franklin Lakes, NJ, USA) and subcutaneously injected into the rabbit 4 times at 1-week intervals. Blood samples were harvested 10 days after the last immunization, and the serum was isolated. Anti-CPS (anti-*S. suis* serotype 2) serum was purchased from SSI Diagnostica (Hillerød, Denmark) and purified by absorption with V5 by Ni-NTA agarose (Qiagen, catalog no. 30210) according to the manufacturer's instructions. Briefly, 200 μ L of V5 (His-tagged, 1 μ g/ μ L) were loaded on the 5 mL Ni-NTA agarose column. The collected antiserum was further purified by incubation with formalin-inactivated Δ 2BSS2 (5×10^{10} CFU) overnight at 4°C. The purified anti-CPS recognized 05ZYH33 but not Δ 2BSS2 as determined by enzyme-linked immunosorbent assay (ELISA) against 05ZYH33 and Δ 2BSS2, respectively. Titers of the purified antisera against V5 and CPS were evaluated by ELISA and stored at -80°C until use.

Passive antiserum transfer

Female C57BL/6J Cnc mice (6–8 weeks) were transferred with anti-CPS or anti-V5 serum through the i.n. or i.v. route. Optimal amounts of antisera were predetermined in mouse infection assays (the minimum amount of anti-V5 to inhibit 05ZYH33 colonization in NALT and the minimum amount of anti-CPS to prevent bacteremia). For anti-CPS serum, 20 μ g/200 μ L (100 μ g/mL) was used for tail vein injection or 5 μ g/30 μ L (170 μ g/mL) for nasal drip. For anti-V5 serum, 100 μ g/200 μ L (500 μ g/mL) and 50 μ g/30 μ L (170 μ g/mL) were used through the i.v. and i.n. routes, respectively. Six hours after antiserum transfer, mice were inoculated via the i.n. route with 2×10^8 CFU of 05ZYH33, and CFUs in different compartments of the infected mice were determined.

Immunofluorescence staining and confocal microscopy

Immunofluorescence staining and confocal microscopy of the nasal cavity were performed as described by Pagelow et al. [29]. Briefly, whole heads of the mice were fixed with 4% PFA for at least 24 h before embedding in paraffin wax. Sections of longitudinally bisected heads were deparaffinized followed by antigen retrieval. Immunostaining was conducted using a primary rabbit anti-*S. suis* antibody (1:100) and a mouse anti-tubulin beta III antibody (Merck Millipore, Darmstadt, Germany catalog no. mab5564, 1:300) to detect neuronal structures, and then appropriate fluorophore-conjugated secondary antibodies: donkey anti-rabbit Cy3 (Jackson Immuno-Research Laboratories, Ely, United Kingdom, catalog no. 711-166-152, 1:1000), donkey anti-mouse FITC (Jackson ImmunoResearch Laboratories, Ely, United Kingdom, catalog no. 715-096-150, 1:500). Tissues were mounted with Vectashield Mounting Medium with DAPI (Vector Laboratories, Burlingame, CA, USA. catalog no. H-1200). Images were captured using a Leica Aperio Versa 200 scanning microscope and color-balanced uniformly across the field of view using the Image Processing Leica Confocal and ImageJ Software (Wayne Rasband, National Institutes of Health, Bethesda, MD, USA).

Histological analysis of brain tissue

Mice were treated with 5 μ L of 1% acetic acid (pH 4.0) per nostril. One hour later, one group of mice was inoculated i.n. with 05ZYH33 (2×10^8 CFU). A control group of mice was given PBS via the i.n. route. Infected mice were euthanized 1 h and 9 days after infection, respectively. The brain was removed and quickly fixed with 4% paraformaldehyde. After paraffin embedding, tissue sections were stained with hematoxylin and eosin (H&E) according to a standard protocol and examined using a Leica SCN400 slide scanner (Leica Microsystems, Germany).

Statistical analysis

Statistical analyses were performed in GraphPad Prism software, version 7.0. CFUs were analyzed by two-tailed unpaired Mann–Whitney U nonparametric *t*-tests. An unpaired, two-tailed Student's *t*-test was used to analyze the statistical significance between two groups. One-way analysis of variance (ANOVA) with Tukey's post-hoc test was used to analyze the statistical significance of differences among more than two groups. A *P*-value of <0.05 was considered significant.

Acknowledgements

The authors acknowledge Prof. Jiaqi Tang of General Hospital of the Eastern Theater of the People's Liberation Army of China for providing the capsular mutant of *S. suis* 05ZYH33 (Δ 2BSS2), Dr. Shengbo Cao (National Key Laboratory of Agricultural Microbiology, Huazhong Agricultural University, Wuhan, China) for sharing *S. suis* serotype strains, Prof. Xiang-Ru Wang in Huazhong Agricultural University, Wuhan, Hubei, China for the human brain microvascular endothelial cells (hBMEC), and Dr. Junfeng Hao in Pathological Laboratory of Institute of Biophysics, CAS, for performing the immunofluorescence microscopy. This research was supported by the National Natural Science Foundation of China grant 31872743 (to B.W.) and the China Postdoctoral Science Foundation Grant 2019M650875 (to X. W.).

Additional information

Author contributions

B.W. conceived the original idea, designed the experiments and wrote the manuscript. X.W. performed the experiments, analyzed the data, and drafted the manuscript. J.W., N.L., and X.F. contributed to mouse experiments and provided advice on experimental designs.

Additional files

Supplementary Table 1. [↗](#)

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Reviewer #1 (Public review):

Summary:

The manuscript by Wang et al. investigates the relationship between *Streptococcus Suis* (S. Suis) growth phases and levels of virulence factor, capsular polysaccharide (CPS), in the bacterial cell wall. They use an understudied mouse intranasal infection model to connect growth phase related CPS abundance to the pathogenicity of the bacteria in the nose, blood, and other organs. Adoptive transfer of serum against either CPS or V5 (five other virulence factors) reinforces their discovery of CPS levels on S. Suis in different organs and stages of infection. Vaccination against bacterial infections can be difficult, and understanding how the serotype of a bacterial pathogen changes between infection sights and systemic disease is critical. Further, understanding host-pathogen interactions at early time points in the upper respiratory tract may have broad implications for vaccine development. While some of the results are interesting and compelling, others are not supported by the data and require further experimental work.

Strengths:

The model of intranasal infection is compelling to expand upon work previously done in vitro and with systemic routes of infection. The histology and fluorescent imaging of the olfactory epithelium and olfactory bulb complement work in Figure 2 about the attachment of S. suis to epithelial cells and the bacterial burden over time in different organs of Figure 3. Histology was performed at 1 hour and 9 days after intranasal infection with stationary phase S. suis and drives home that this pathogen can invade the olfactory nerve and may potentially cause bacterial meningitis seen in some infected swine.

The adoptive transfer of either anti-CPS or anti-V5 to mice before infection at both longer (12 hr), and shorter (0.5 hr) time points is useful to demonstrate that the changes in cell wall composition between the NALT/CSF and blood compartments result in different efficacy in clearing bacteria from those locations. This is fundamental for the development of vaccines for the swine industry and begs those developing other bacterial vaccines to consider what virulence factors are the most useful as neutralizing antibody targets at the sight of bacterial invasion.

Demonstrating that the amount of CPS within the cell wall of *S. Suis* is related to the growth phase of the bacteria is an important consideration for vaccine development. While others had previously shown that CPS levels were higher in the blood than in the CNS, and that CPS decreases the invasion of epithelial cells, the close look at the olfactory epithelium at an early time point ties together in vitro findings. The control of a CPS-negative strain was critical to understanding their findings. The location and the microbial community that bacterial pathogens live within may change the growth phase and therefore also the cell wall components.

Weaknesses:

The authors present compelling data that is relevant to the development of anti-bacterial vaccinations and show a relationship between CPS levels and pathogenicity. However, the use of a laboratory murine model requiring acetic acid pre-treatment and a high i.n. dose. Therefore, the findings presented may not represent what occurs in swine. Furthermore, several conclusions are not supported by the data and require substantial new experimental support. Thus, major concerns remain that impact the validity of the findings.

Major concerns for the manuscript:

The intranasal infections were done with *S. Suis* in the stationary phase which has been shown to have less CPS on the cell wall. While this mimics the literature that shows *S. Suis* to have less CPS in the CNS, the difference in the pathogenesis of a log phase vs. stationary phase intranasal infection would be interesting. Especially because the bacteria is a part of the natural microbial community of swine tonsils, it is curious if the change in growth phase and therefore CPS levels may be a causative reason for pathogenic invasion in some pigs. To take this line of thought a step further, the authors should consider taking the bacteria from NALT/CSF and blood and compare the lag times bacteria from different organs take to enter a log growth phase to show whether the difference in CPS is because *S. Suis* in each location is in a different growth phase. If log phase bacteria were intranasally delivered, would it adapt a stationary phase life strategy? How long would that take? Lastly, the authors should be cautious about claims about *S. suis* downregulating CPS in the NALT for increased invasion and upregulating CPS to survive phagocytosis in blood. While it is true that the data shows that there are different levels of CPS in these locations, the regulation and mechanism of the recorded and observed cell wall difference is not investigated past the correlation to the growth phase. While mechanistic work is outside the scope of the current work, readers should keep in mind that these results may be explained multiple ways. In addition, the mouse model is used rather than the usual host of a pig. The NALTs of conventional pigs and SPF mice certainly have unique microbial communities and this may affect the pathogenesis of *S. suis* in the mouse, therefore influencing the results. Because the authors show a higher infection rate in the mouse with acetic acid, they may want to consider investigating what the mouse NALT microenvironment is naturally doing to exclude more bacterial invasion in future studies. Is it simply a host mismatch or is there something about the microbiome or steady-state immune system in the nose of mice that is different from pigs?

<https://doi.org/10.7554/eLife.101760.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer 1:

(1) Some conclusions are not completely supported by the present data, and at times the manuscript is disjoint and hard to follow. While the work has some interesting observations, additional experiments and controls are warranted to support the claims of the manuscript.

Thank you for the comments. We revised some of the claims and conclusions to be more objective and result-supportive.

(2) While the authors present compelling data that is relevant to the development of anti-bacterial vaccinations, the data does not completely match their assertions and there are places where some further investigation would further the impact of their interesting study.

We do not fully agree with the reviewer's comments. We have demonstrated that changes in CPS levels during infection are associated with pathogenesis, which will guide future studies on the underlying mechanisms. A significant amount of effort is required for studying mechanisms, which is beyond the scope of this research. We concur with the reviewer that assertions should be made cautiously until further studies are conducted. We have revised these assertions to align with the data and to avoid extrapolating the results (pages 7, lines 126, 133-136; page 11, lines 216-218; page 13, line 264; and page 18, lines 378-383).

(3) The difference in the pathogenesis of a log phase vs. stationary phase intranasal infection would be interesting. Especially because the bacteria is a part of the natural microbial community of swine tonsils, it is curious if the change in growth phase and therefore CPS levels may be a causative reason for pathogenic invasion in some pigs.

S. suis is a part of the natural microbial community of swine tonsils but not mouse NALT. It is interesting to know if CPS levels are low in pig tonsils since CPS is hydrophilic and not conducive to bacterial adhesion. In the study, mice were i.n. infected with a high dose of the bacteria, which could increase opportunities for dissemination (acidic acid may not be a contributor since with or without it is similar). *S. suis* getting into other body compartments from pig tonsils might be triggered by other conditions, such as viral coinfection, nasal cavity inflammation, cold weather, and decreased immunity.

Experiments with pig blood and phagocytes have shown that genes involved in the synthesis of CPS are upregulated in pig blood. In contrast, these genes are downregulated [1]. In addition, the absence of CPS correlated with increased hydrophobicity and phagocytosis, proposing that *S. suis* undergoes CPS phase variation and could play a role in the different steps of *S. suis* infection [2]. We showed direct evidence of encapsulation modulation associated with *S. suis* pathogenesis in mice. A pig infection model is required to confirm these findings.

(4) The authors should consider taking the bacteria from NALT/CSF and blood and compare the lag times bacteria from different organs take to enter a log growth phase to show whether the difference in CPS is because S. suis in each location is in a different growth phase. If log phase bacteria were intranasally delivered, would it adapt a stationary phase life strategy? How long would that take?

What causes CPS regulation in vivo is not known. CPS changes in different culture stages, indicating that stress, such as nutrition levels, is one of the signals triggering CPS regulation. The microenvironment in the body compartments is far more complex than in vitro, in which host cells, immune factors and others may affect CPS regulation, individually or collectively.

The reviewer's question is important but the suggested experiment is impracticable since bacterial numbers taken from organs are few, and culturing the bacteria in vitro would obliterate the in vivo status.

(5) Authors should be cautious about claims about S. suis downregulating CPS in the NALT for increased invasion and upregulating CPS to survive phagocytosis in blood. While it is true that the data shows that there are different levels of CPS in these locations, the regulation and mechanism of the recorded and observed cell wall difference are not investigated past the correlation to the growth phase.

We lower the tone and change the claim as “suggest a correlation between lower CPS in the NALT and a greater capacity for cellular association, whereas elevated CPS levels in the blood are linked to improved resistance against bactericidal activity. However, the mechanisms behind these associations remain unknown.” (page 7, lines 133-136).

(6) The mouse model used in this manuscript is useful but cannot reproduce the nasal environment of the natural pig host. It is not clear if the NALTs of pigs and mice have similar microbial communities and how this may affect the pathogenesis of S. Suis in the mouse. Because the authors show a higher infection rate in the mouse with acetic acid, they may want to consider investigating what the mouse NALT microenvironment is naturally doing to exclude more bacterial invasion. Is it simply a host mismatch or is there something about the microbiome or steady-state immune system in the nose of mice that is different from pigs?

It is a very interesting comment. The mice are SPF level. The microenvironment in SPF mouse NALT should be significantly different from conventional pig tonsils. Although NALT in mice resembles pig tonsils in function, many factors may contribute to the sensitivity to *S. suis* colonization in the pig nasal cavity, such as the microbiome and local steady-state immune system. More complex microbiota in tonsils could be one of the factors. Analyzing what makes *S. suis* inclined towards colonization in pig tonsils by SPF and conventional pigs are an ideal experiment to answer the question.

(7) Have some concerns regarding the images shown for neuroinvasion because I think the authors mistake several compartments of the mouse nasal cavity as well as the olfactory bulb. These issues are critical because neuroinvasion is one of the major conclusions of this work.

Thank you for your comments. The olfactory epithelium (OE) is located directly underneath the olfactory bulb in the olfactory mucosa area and lines approximately half of the nasal cavities of the nasal cavity. The remaining surface of the nasal cavity is lined by respiratory epithelium, which lacks neurons. The olfactory receptor neuron in OE is stained green in the images by β -tubulin III, a neuron-specific marker. The respiratory epithelium is colorless due to the absence of nerve cells. Similarly, the green color stained by β -tubulin III identifies the olfactory bulb. The accuracy of the anatomic compartments of the mouse nasal cavity has been checked and confirmed by referring to related literature [3, 4].

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(3) Pögelow D, Chhatbar C, Beineke A, Liu X, Nerlich A, van Vorst K, Rohde M, Kalinke U, Förster R, Halle S, Valentin-Weigand P, Hornef MW, Fulde M. The olfactory epithelium as a port of entry in neonatal neurolisterosis. *Nat Commun*. 2018;9(1):4269.

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Reviewer 2:

(1) However, there are serious concerns about data collection and interpretation that require further data to provide an accurate conclusion. Some of these concerns are highlighted below:

Both reviewers were concerned about some of the interpretations of the results. We modified the interpretations in related lines throughout the manuscript (Please see the related responses to Reviewer 1).

*(2) In figure 2, the authors conclude that high levels of CPS confer resistance to phagocytic killing in blood exposed *S. suis*. However, it seems equally likely that this is resistance against complement mediated killing. It would be important to compare *S. suis* killing in animals depleted of complement components (C3 and C5-9).*

We thank the reviewer for the comment. The experiment should be Bactericidal Assay instead of anti-phagocytosis killing. CPS is a main inhibitor of C3b deposition [1]. It interferes with complement-mediated and receptor-mediated phagocytosis; and direct killing. Data in Figure 2C is expressed as “% of bacterial survival in whole blood” for clarity (page 8, Fig. 2C and page 23, lines 489-490).

(3) Intranasal administration non-CPS antisera provides a nice contrast to intravenous administration, especially in light of the recently identified "blood-olfactory barrier". Can the authors provide any insight into how long and where this antibody would be located after intranasal administration? Would this be antibody mediated cellular resistance, or something akin to simple antibody "neutralization"

Anti-V5 may not stay long locally following intranasal administration. Efficient reduction of *S. suis* colonization in NALT supports that anti-V5 could recognize and neutralize the bacteria in NALT quickly, thereby reducing further dissemination in the body. Antibody-mediated phagocytosis may not play a major role because neutrophils are mainly present in the blood but not in the tissues.

(4) The micrographs in Figure 7 depict anatomy from the respiratory mucosa. While there is no histochemical identification of neurons, the tissues labeled OE are almost certainly not olfactory and in fact respiratory. However, more troubling is that in figures 7A,a,b,e, and f, the lateral nasal organ has been labeled as the olfactory bulb. This undermines the conclusion of CNS invasion, and also draws into question other experiments in which the brain and CSF are measured.

We understand the significance of your concerns and appreciate your careful review of Figure 7. The olfactory epithelium (OE) is situated directly beneath the olfactory bulb in the olfactory mucosa area and covers about half of the nasal cavity. This positioning allows information transduction between the olfactory and the olfactory epithelium. The remaining surface of the nasal cavity is lined with respiratory epithelium, which does not contain neurons and primarily serves as a protective barrier. In contrast, the olfactory epithelium consists of basal cells, sustentacular cells, and olfactory receptor neurons. The olfactory

receptor neurons are specifically stained green in the images using β -tubulin III, a marker that is unique to neurons. The respiratory epithelium appears colorless due to the lack of nerve cells. Similarly, the green staining with β -tubulin III also highlights the olfactory bulb. The anatomical structures indicated in the images are consistent with those described in the literature [2, 3], confirming that the anatomy of the nasal cavity has been accurately identified.

(5) Micrographs of brain tissue in 7B are taken from distal parts of the brain, whereas if olfactory neuroinvasion were occurring, the bacteria would be expected to arrive in the olfactory bulb. It's also difficult to understand how an inflammatory process would be developed to this point in the brain -even if we were looking at the appropriate region of the brain -within an hour of inoculation (is there a control for acetic acid induced brain inflammation?). Some explanations about the speed of the immune responses recorded are warranted.

Thank you for highlighting this issue. Cerebrospinal fluid (CSF) flows into the subarachnoid space surrounding the spinal cord and the brain. There are direct connections from this subarachnoid space to lymphatic vessels that wrap around the olfactory nerves as they cross the cribriform plate towards the nasal submucosa. This connection allows for the drainage of CSF into the nasal submucosal lymphatics in mice [4, 5]. Bacteria may utilize this CSF outflow channel in the opposite direction, which explains the development of brain inflammation in the distal areas of brain tissue adjacent to the subarachnoid space. We have included additional relevant information in the revised manuscript (page 16, lines 323-325).

(6) The detected presence of S. suis in the CSF 0.5hr following intranasal inoculation is difficult to understand from an anatomical perspective. This is especially true when the amount of S. suis is nearly the same as that found within the NALT. Even motile pathogens would need far longer than 0.5hr to get into the brain, so it's exceedingly difficult to understand how this could occur so extensively in under an hour. The authors are quantifying CSF as anything that comes out of the brain after mincing. Firstly, this should more accurately be referred to as "brain", not CSF. Secondly, is it possible that the lateral nasal organ -which is mistakenly identified as olfactory bulb in figure 7- is being included in the CNS processing? This would explain the equivalent amounts of S. suis in NALT and "CSF".

The high dose of inoculation used in the experiment may explain the rapid presence of *S. suis* in the CSF. Mice exhibit low sensitivity to *S. suis* infection, and the range for the effective intranasal infectious dose is quite narrow. Higher doses lead to the quick death of the mice, while lower doses do not initiate an infection at all. The dose used in this study is empirical and is intended to facilitate the observation of the progression of *S. suis* infection in mice.

The NALT tissue and CSF samples are collected separately. After obtaining the NALT tissue, the nasal portion was carefully separated from the rest of the head along the line of the eyeballs. The brain tissue was then extracted from the remaining part of the head to collect the CSF, and it was lacerated to expose the subarachnoid space without being minced. This procedure aims to preserve the integrity of the brain tissue as much as possible. Further details about the CSF collection process can be found in the Materials and Methods section (page 24, lines 508-512).

(7) To support their conclusions about neuroinvasion along the olfactory route and /CSF titer the authors should provide more compelling images to support this conclusion: sections stained for neurons and S. suis, images of the actual olfactory bulb (neurons, glomerular structure etc).

Thank you. We respectfully disagree with the reviewer. We stained neurons using a neuron-specific marker to identify the anatomical structures of the olfactory bulb and olfactory epithelium (in green). We used an *S. suis*-specific antibody to highlight the bacteria present in these areas (in orange and red). The images, along with the bacteria found in the cerebrospinal fluid (CSF) and the brain inflammation observed early in the infection, strongly support our conclusion regarding brain invasion through the olfactory pathway. Please see the response to question 4 for further clarification.

References

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Response to Recommendations for the authors:

Reviewer 1:

Minor concerns for the manuscript:

(1) *In the introduction, please consider giving a little more background about the bacteria itself and how it causes pathogenesis.*

We appreciate your suggestion. We have included additional background on the virulent factors and the pathogenesis of the bacteria in the introduction to enhance understanding of the results (page 4, lines 63-69).

(2) *Figure 2C would be more correct to say percent survival as the CFUs before and after are what are being compared and not if the bacteria is being phagocytosed or not. Flow cytometry of the leukocytes and a fluorescent *S. Suis* would show phagocytosis. Unless that experiment is performed, the authors cannot claim that there is a resistance to phagocytosis.*

Thank you for your feedback. We agree with the reviewer that the experiment should be Bactericidal Assay rather than anti-phagocytosis killing. CPS interferes with complement-mediated phagocytosis and direct killing, and receptor-mediated phagocytosis. To enhance clarity, the data in Fig. 2C has been presented as “% of bacterial survival in whole blood” (page 8).

(3) *There are two different legends present for Figure 1. Please resolve.*

We apologize for the oversight. The redundant figure legend has been removed (page 6).

(4) There are places such as in lines 194-195, that there are assertions and interpretations about the data that are not directly drawn from the data. These hypotheses are valuable, but please move them to the discussion.

Thank you for your suggestion. The hypothesis has been moved to the Discussion section (page 19, lines 402 - 405).

(5) In Figure 4B, higher resolution images would strengthen the ability of non-microbiologists to see the differences in CPS levels in the cell wall.

We achieved the highest resolution possible for clearer distinctions in CPS levels. To enhance the visualization of the different CPS levels in the images, we revised the description of the CPS changes in Figure 4B within the results section (page 11, lines 208-213).

(6) In Figure 5 there is no D. Further, the schematics throughout would be easier to parse with the text if the challenge occurred at time 0. Consider revising them for clarity.

Thank you for highlighting the error. We have removed "i.v + i.n (Fig. 5)" from Figure 5A and made adjustments to the schematic illustrations in Figures 5 and 6 as recommended by the reviewer (page 14).

(7) What is the control for the serum? The findings for figures 5 and 6 would be much stronger if a non- S. Suis isotype control serum was also infused.

We used a naive serum as a control to avoid interference from a non-S. suis isotype control that targets other surface molecules of S. suis serotypes.

(8) Figure 6 legend does not include the anti-CPS treatment.

Thank you. We have added anti-CPS serum in the legend (page 15, line 249).

(9) Figure 7 legend does not include the time point for panel 7A.

Thank you. The time point is shown on Fig.7A (page 17).

(10) Figure 7 should show OB micrographs or entire brain including the OB.

The neuron-specific marker, β -tubulin III, identifies the neuro cells in the olfactory bulb (OB) as shown in Fig. 7A. Unfortunately, we were unable to provide an image of the entire brain that includes the OB due to limitations in our section preparation. We apologize for the mislabeled structure in Fig. 7A, which may have caused confusion. We have corrected the labeling for consistency (see page 15, lines 257-260). Additionally, we included a drawing of the sagittal plane of the rodent's nose, depicting the compartments of the OB, olfactory epithelium (OE), nasal cavity (NC), and brain. This illustration, presented in Fig. 7B on page 17, aims to clarify the structural and functional connections between the nasopharynx and the CNS.

(11) Some conclusions may be better drawn if figures were to be consolidated. As noted above, the data at times feels disjointed and the importance is more difficult for readers to follow because data are presented further apart. Particularly figures 5 and 6 which are similar with different time points and controls of antisera administrative routes;

placing these figures together would be an example of increasing continuity throughout the paper.

Thank you for the valuable suggestion. Figures 5 and 6, along with their related descriptions in the results section, have been combined for better cohesiveness (pages 14-15).

Reviewer #2:

To support their conclusions about neuroinvasion along the olfactory route and /CSF titer the authors should provide more compelling images to support this conclusion: sections stained for neurons and S. suis, images of the actual olfactory bulb (neurons, glomerular structure etc).

Please refer to our responses to Reviewer 1's Question 7, Reviewer 2's Questions 4 and 7 in the public reviews, and Reviewer 1's Question 10 in the authors' recommendations.

<https://doi.org/10.7554/eLife.101760.2.sa0>